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Interactive Applications Using Matplotlib

Don't just see your data, experience it!

Benjamin V. Root

[PACKT] open source*
PUBLISHING community experience distilled

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BIRMINGHAM - MUMBAI

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Benjamin V. Root has been a member of the Matplotlib development team since 2010. His main areas of development have been the documentation and the mplot3d toolkit, but now he focuses on code reviews and debugging. Ben is also an active member of mailing lists, using his expertise to help newcomers understand Matplotlib. He is a meteorology graduate student, working part-time on his PhD dissertation. He works full-time for Atmospheric and Environmental Research, Inc. as a scientific programmer.

I would like to acknowledge the entire Matplotlib development team for their insightful responses to my questions while I was writing this book. In particular, I would like to thank Michael Droettboom, Eric Firing, Thomas Caswell, Phil Elson, and Ryan May. Thanks also go to the members of the matplotlib users' list without whom I would have never learned this tool in the first place and for whom I wrote this book.

This book would not have been possible without the love and support of my wife, Margaret. She put up with far more than she should have, and for that, I am in her debt.

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During his master's degree in telecommunications engineering, he engaged in part-time research in the Department of Signal Theory and Communications at the Polytechnic University of Catalonia (UPC), with the design and development of a low-cost hyperspectral in-situ sensor. This experience stimulated him to start a PhD at the same department. He obtained a grant from the Spanish National Research Council (CSIC) and performed his predoctoral training at the Marine Technology Unit in Barcelona, graduating for a master of advanced studies and leading and supervising the master thesis of other university students, while continuing his research on low-cost solutions oriented to increase the observational capabilities for marine/oceanographic biological information systems.

In 2011, he gained a fellowship from the Spanish Ministry of Economy and Competitiveness to expand his experience in international scientific organisms, moving to the European Space Agency office in Italy and working on assessing the viability of remote sensing coral monitoring. During his stay, he gained a contractor position as performance simulation engineer for the Sentinel 3 satellite project at the European Space Agency facilities in the Netherlands, being responsible for the simulators and processors operation and maintenance.

In January 2015, he moved to San Diego, California, where he is currently finishing his PhD while he pursues new opportunities.

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Preface

Why Matplotlib? Why Python, for that matter? I picked up Python for scientific development because I needed a full-fledged programming language that made sense. Too often, I felt hemmed in by the traditional tools in the meteorology field. I needed a language that respected my time as a developer and didn't fight me every step of the way. "Don't you find Python constricting?" asked a colleague who was fond of bad puns. "No, quite the opposite," I replied, the joke going right over my head.

Matplotlib is the same in this respect. Switching from traditional graphing tools of the meteorology field to Matplotlib was a breath of fresh air. Not only were useful programs being written using the Matplotlib library, but it was also easy to write my own. Furthermore, I could write out modules and easily use them in both the hardcopy generating scripts for my publications and for my data exploration interactive applications. Most importantly, the Matplotlib library let me do what I needed it to do.

I have been an active developer for Matplotlib since 2010 and I am *still* discovering Matplotlib. It isn't that the library is insanely huge and unwieldy — it isn't. Instead, Matplotlib appeals to all levels of expertise and interests. One can simply care enough only to get a single plot displayed in three line of code and never think of the library again. Or, one could assume control over every single minute plotting detail, ensuring that everything is displayed "just right." And even when one does this and thinks they have seen every single nook and cranny of the library, they will discover some other feature that they have never seen before.

Matplotlib is 12 years old now. New plotting projects have cropped up — some supplementing Matplotlib's design, while others trying to replace Matplotlib entirely. However, there has been no slacking of interest in Matplotlib, not from the users and definitely not from the developers. The new projects are interesting, and as with all things open source, we try to learn from these projects. But I keep coming back to this project. Its design, developers, and community of users are some of the best and most devoted in the open source world.

The book you are reading right now is actually not the book I originally wanted to write. The interactive aspect of Matplotlib is not my area of expertise. After some nudging from fellow developers and users, I relented. I proceeded to rewrite the only interactive application I had ever finished and published. Working through the chapters, I tried to find better ways of doing the things I did originally, pointing out major pitfalls and easy mistakes as I encountered them. It was a significant learning experience for me, which was wholly unexpected.

I now invite you to discover Matplotlib for yourself. Whether it is the first time or not, it certainly won't be the last.

What this book covers

Chapter 1, Introducing Interactive Plotting, covers basic figure-axes-artist hierarchy and other Matplotlib essentials such as displaying the plot. It also introduces you to the interactive Matplotlib figure.

Chapter 2, Using Events and Callbacks, provides Matplotlib's events and a callback system to bring your figures to life. It also explains how you can extend it with custom events, making the application truly interactive.

Chapter 3, Animations, deals with `ArtistAnimation`, `FuncAnimation`, and timers to make animations of all types. It also deals with animations that can be saved as movies.

Chapter 4, Widgets, covers built-in widgets such as buttons, checkboxes, selectors, lassos, and sliders, which are all explained and demonstrated. Here, you'll also learn about other useful third-party widgets and tools.

Chapter 5, Embedding Matplotlib, teaches you how to add GUI elements to an existing Matplotlib application. Here you'll also see how to add your interactive Matplotlib figure to an existing GUI application. Identical examples are presented using GTK, Tkinter, wxWidgets, and Qt.

What you need for this book

At the absolute least, you will need the following Python packages installed on your system: NumPy, SciPy, Basemap, and (of course) Matplotlib. To work on the instructions presented in *Chapter 5, Embedding Matplotlib*, you will want to have at least one of the following GUI toolkits installed: GTK, Tkinter (should come with Python), wxWidgets, or Qt (version 4 is preferred; version 5 is supported only recently for Matplotlib version 1.4). You will also need the corresponding Python bindings for the GUI toolkits (some come with them by default).

Who this book is for

If you are a Python programmer who wants to do more than just see your data, this is the book for you. It will explain the SciPy stack (that is, NumPy and Matplotlib) and provide pointers to install them. Experience with GUI toolkits, such as wxPython, Qt, or GTK+, is also not required, so this book can be an excellent complement to other GUI programming resources. To understand the examples and explanations, you need to know basic object-oriented programming terms and concepts.

Conventions

In this book, you will find a number of text styles that distinguish between different kinds of information. Here are some examples of these styles and an explanation of their meaning.

Code words in text, database table names, folder names, filenames, file extensions, pathnames, dummy URLs, user input, and Twitter handles are shown as follows: "We can include other contexts through the use of the `include` directive."

A block of code is set as follows:

```
import matplotlib.pyplot as plt
from matplotlib.collections import LineCollection
from tutorial import track_loader
tracks = track_loader('polygons.shp')
# Filter out non-tracks (unassociated polygons given trackID of -9)
tracks = {tid: t for tid, t in tracks.items() if tid != -9}
```

When we wish to draw your attention to a particular part of a code block, the relevant lines or items are set in bold:

```
polys = [p for p in cells.polygons]
for p in polys:
    p.set_visible(True)
    p.set_alpha(0.0)

def update(frame, polys):
    for i, p in enumerate(polys):
        alpha = 0.0 if i > frame else 1.0 / ((frame - i + 1)**2)
        p.set_alpha(alpha)

ax.set_xlabel("Longitude")
ax.set_ylabel("Latitude")
strmanim = FuncAnimation(fig, update, frameCnt,
                        fargs=(polys,))

plt.show()
```

Any command-line input or output is written as follows:

```
$ pip install matplotlib
```

New terms and **important words** are shown in bold. Words that you see on the screen, for example, in menus or dialog boxes, appear in the text like this: "Now click on the **Selection** radio button and you will find that you can select a polygon again."

 Warnings or important notes appear in a box like this.

 Tips and tricks appear like this.

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